

langevin-lean: Formal Proofs of Bounded-Noise Langevin Convergence in Lean 4

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Abstract

`langevin-lean` is a Lean 4 / Mathlib library formalising a deterministic bounded-noise convergence argument for Langevin dynamics. The library works over a real inner product space, packages an abstract gradient map with Lipschitz and strong-monotonicity hypotheses, and studies the noisy update $\mathbf{x}_{\{k+1\}} = \mathbf{x}_k - \alpha \text{grad}_f(\mathbf{x}_k) + \mathbf{x}_i_k$. It proves gradient-step contraction, a one-step contraction-plus-noise bound, a linear recurrence estimate, and geometric convergence to a noise-dependent ball. The formal development is machine-checked in Lean 4 with zero `sorry`, zero `admit`, and standard Lean/Mathlib axioms only.

1. Introduction

The overdamped Langevin stochastic differential equation is a basic model for sampling from densities proportional to $\exp(-f)$. Its Euler discretisation, the unadjusted Langevin algorithm, combines a gradient drift term with Gaussian noise. Classical convergence results for this process are distributional: they are stated in Wasserstein distance, KL divergence, or total variation, and they rely on stochastic calculus and measure-theoretic functional inequalities.

`langevin-lean` formalises a deterministic proxy for that analysis. The Gaussian perturbation is replaced by a bounded disturbance, and the proof follows the squared distance to the minimiser. The resulting theorem is a contraction-plus-noise statement: the iterates converge geometrically to a ball whose radius is determined by the noise magnitude. This is not a full stochastic Langevin theorem, but it captures the finite-dimensional Lyapunov calculation that underlies many stability proofs for noisy gradient systems.

The repository is deliberately focused. It does not formalise Ito integration, Girsanov transformations, log-Sobolev inequalities, or distributional convergence. Instead, it gives a compact, importable Lean 4 proof of the deterministic recurrence argument under explicit smoothness, strong convexity, step-size, and bounded-noise hypotheses.

2. Library Overview

The project is organised into three implementation modules plus a root import file:

- `LangevinLean/Defs.lean` defines `LangevinSetup`, the contraction factor `rho`, the update `step`, the iterate sequence, the effective contraction `rho_eff`, the linear recurrence constant, and the final noise radius.
- `LangevinLean/StepBound.lean` proves the gradient norm bound, the strong-convexity inner-product bound, deterministic gradient-step contraction, and the one-step noisy bound.
- `LangevinLean/Convergence.lean` proves the contraction condition, non-negativity of `rho`, the linear recurrence estimate, Young’s inequality, cross-term absorption, and the final convergence theorem.
- `LangevinLean.lean` is the root module importing the library.

The project depends on:

- Lean v4.28.0
- Mathlib v4.28.0

The formal development contains zero `sorry`, zero `admit`, and introduces no project-specific axioms. It is written against Lean 4 and Mathlib, using standard Lean/Mathlib axioms only.

The formal setting is a real inner product space:

```
variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace Real E]
```

A `LangevinSetup` packages:

```
grad_f : E -> E
x_star : E
L, m, alpha, sigma : Real
with hypotheses
0 < L, 0 < m, 0 < alpha, 0 <= sigma, m <= L
grad_f x_star = 0
||grad_f x - grad_f y|| <= L * ||x - y||
m * ||x - y||^2 <= <grad_f x - grad_f y, x - y>.
```

The repository is available at:

<https://github.com/velvetmonkey/langevin-lean>

3. Theorem Inventory

The source contains sixteen headline theorem-level results.

Layer 1 - One-Step Geometry

1. `grad_norm_bound` proves that Lipschitzness and `grad_f x_star = 0` control the gradient norm:

```
||grad_f x|| <= L * ||x - x_star||.
```

2. `inner_grad_lower_bound` derives the strong-convexity inner-product lower bound:

$$m * ||x - x_{\text{star}}||^2 \leq \langle \text{grad}_f x, x - x_{\text{star}} \rangle.$$

3. `gradient_step_sq_bound` proves deterministic gradient-step contraction:

$$||x - \alpha * \text{grad}_f x - x_{\text{star}}||^2 \leq \rho * ||x - x_{\text{star}}||^2.$$

4. `gradient_step_norm_le` proves that if $\rho \leq 1$, then the deterministic gradient step does not increase distance to the minimiser.

5. `langevin_step_bound` proves the one-step bounded-noise estimate:

$$\begin{aligned} ||\text{step } x \text{ xi} - x_{\text{star}}||^2 &\leq \rho * ||x - x_{\text{star}}||^2 \\ &\quad + 2 * (\alpha * \sigma) * ||x - x_{\text{star}}|| \\ &\quad + 2 * (\alpha * \sigma)^2. \end{aligned}$$

Layer 2 - Contraction and Recurrences

6. `contraction_condition` proves the sufficient step-size condition:

$$\alpha < 2 * m / L^2 \rightarrow \rho < 1.$$

7. `rho_le_one` proves the weak version:

$$\alpha \leq 2 * m / L^2 \rightarrow \rho \leq 1.$$

8. `rho_nonneg` proves

$$0 \leq \rho.$$

9. `linear_recurrence_bound` proves the general scalar recurrence theorem:

$$\begin{aligned} a(n+1) &\leq \rho * a(n) + c \\ \rightarrow a(n) &\leq \rho^n * a(0) + c / (1 - \rho), \end{aligned}$$

under $0 \leq \rho < 1$ and $0 \leq c$.

10. `scaled_young` proves the scaled Young inequality:

$$2 * a * b \leq \epsilon * a^2 + b^2 / \epsilon.$$

Layer 3 - Convergence

11. `step_to_linear_recurrence` absorbs the cross term from `langevin_step_bound` into a linear recurrence:

$$\begin{aligned} ||\text{step } x \text{ xi} - x_{\text{star}}||^2 &\leq \rho_{\text{eff}} * ||x - x_{\text{star}}||^2 + c_{\text{linear}}. \end{aligned}$$

12. `rho_eff_nonneg` proves $0 \leq \rho_{\text{eff}}$ when $0 \leq \rho$.

13. `rho_eff_lt_one` proves $\rho_{\text{eff}} < 1$ when $\rho < 1$.

14. `c_linear_nonneg` proves $0 \leq c_linear$ under $\rho < 1$.

15. `noiseRadius_eq` proves the algebraic identity:

$$\text{noiseRadius} = c_linear / (1 - \rho_eff).$$

16. `langevin_convergence` is the final theorem:

$$\begin{aligned} &||\text{iterate } x_0 \text{ noise } k - x_star||^2 \\ &\leq \rho_eff^k * ||x_0 - x_star||^2 + \text{noiseRadius}. \end{aligned}$$

4. Main Theorems

Gradient-Step Contraction

The theorem `gradient_step_sq_bound` expands the squared distance after the deterministic gradient step. The strong-convexity hypothesis controls the descent inner product, and the Lipschitz-gradient hypothesis controls the squared gradient norm. The result is the familiar factor

$$\rho = 1 - 2 * \alpha * m + \alpha^2 * L^2.$$

One-Step Bounded-Noise Bound

The theorem `langevin_step_bound` adds a perturbation `xi` satisfying

$$||xi|| \leq \alpha * \sigma.$$

The proof expands the squared norm of the sum of the deterministic step displacement and the noise. Cauchy-Schwarz and the non-expansion result for $\rho \leq 1$ control the cross term, giving the contraction-plus-noise inequality.

Linearised Recurrence

The theorem `step_to_linear_recurrence` applies Young's inequality to the cross term

$$2 * \alpha * \sigma * ||x - x_star||.$$

With $\epsilon = (1 - \rho) / 2$, this converts the nonlinear one-step estimate into

$$a_{k+1} \leq \rho_eff * a_k + c_linear.$$

Convergence to a Noise Ball

The theorem `langevin_convergence` assumes bounded noise at every step and $\rho < 1$. It proves

$$\begin{aligned} &\text{forall } k, \\ &||S.\text{iterate } x_0 \text{ noise } k - S.x_star||^2 \\ &\leq S.\rho_eff^k * ||x_0 - S.x_star||^2 + S.\text{noiseRadius}. \end{aligned}$$

This is geometric convergence to a deterministic noise-dependent ball.

5. Proof Sketch

`LangevinLean/Defs.lean` sets up the abstract deterministic model. The stochastic Langevin update is represented by an arbitrary noise sequence, with stochastic assumptions replaced by the pointwise deterministic bound $\|\text{noise } k\| \leq \alpha * \sigma$.

`LangevinLean/StepBound.lean` proves the local calculation. Smoothness gives `grad_norm_bound`; strong convexity gives `inner_grad_lower_bound`. Expanding the squared distance after $x - \alpha \text{ grad_f } x$ yields `gradient_step_sq_bound`. Adding noise and applying the real inner-product norm expansion yields `langevin_step_bound`.

`LangevinLean/Convergence.lean` proves the scalar facts needed to iterate the local estimate. The step-size condition implies $\rho < 1$; the assumption $m \leq L$ gives ρ_{nonneg} . Young’s inequality absorbs the cross term, and induction on k proves the resulting linear recurrence. The identity `noiseRadius_eq` rewrites the steady-state term into the final displayed form.

6. Relation to Sibling Libraries

`langevin-lean` is closest to `gradient-descent-lean` and `sgd-lean`. `gradient-descent-lean` has DOI 10.5281/zenodo.20472996 and proves deterministic smooth convex descent. `sgd-lean` has DOI 10.5281/zenodo.20475583 and studies a bounded-noise oracle model for stochastic gradient descent. The present library uses a similar deterministic bounded-noise philosophy, but for the Langevin-style update and a strongly convex contraction analysis.

`contraction-lean` has DOI 10.5281/zenodo.20474762 and formalises convergence through metric contraction. The proof in `langevin-lean` is an affine contraction with an additive disturbance term. `mirror-descent-lean` has DOI 10.5281/zenodo.20475033 and uses a Bregman potential rather than Euclidean squared distance. Together these repositories cover several common proof patterns: exact descent, noisy descent, contraction, and geometric optimisation.

7. Conclusion

`langevin-lean` provides a compact Lean 4 / Mathlib formalisation of a bounded-noise Langevin convergence argument. It defines the deterministic proxy for Langevin dynamics, proves contraction of the gradient step, bounds the effect of bounded perturbations, converts the estimate into a linear recurrence, and proves geometric convergence to a noise-dependent ball.

Future work could connect this deterministic proxy to formal probability theory, stochastic processes, Wasserstein metrics, and log-Sobolev inequalities. Those ingredients would be needed for a full machine-checked theorem about stochastic

Langevin sampling. The current repository supplies the checked contraction-plus-noise core.

References

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